Chapter 7

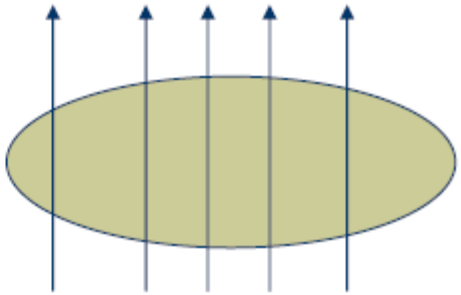
Time-varying Fields and Maxwell's Equations

7.2 Faraday's Experimental Law

$$emf = -\frac{d\Phi}{dt} \quad (\text{V})$$

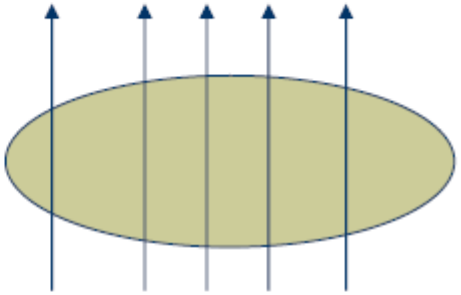
Electromotive force

Rate of change of magnetic flux passing A CLOSED LOOP



Faraday's Law in Integral Form

$$emf = -\frac{d\Phi}{dt} \quad (\mathrm{V})$$



$$emf = \oint_C \vec{E} \cdot d\vec{l}$$

$$-\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}$$

$$\int_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}$$

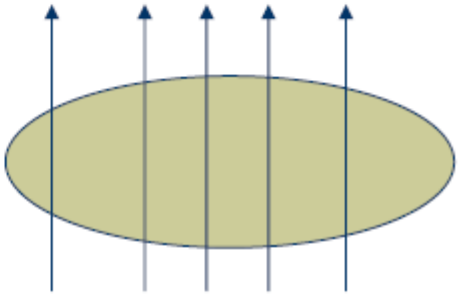
Faraday's Law in Differential Form

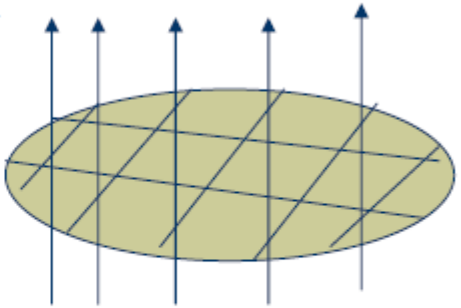
$$\int_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}$$

Stoke's Theorem

$$\oint_C \vec{E} \cdot d\vec{l} = \int_S \nabla \times \vec{E} \cdot d\vec{S} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}$$

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$





At a point

How to generate the electromotive force

$$\int_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}$$

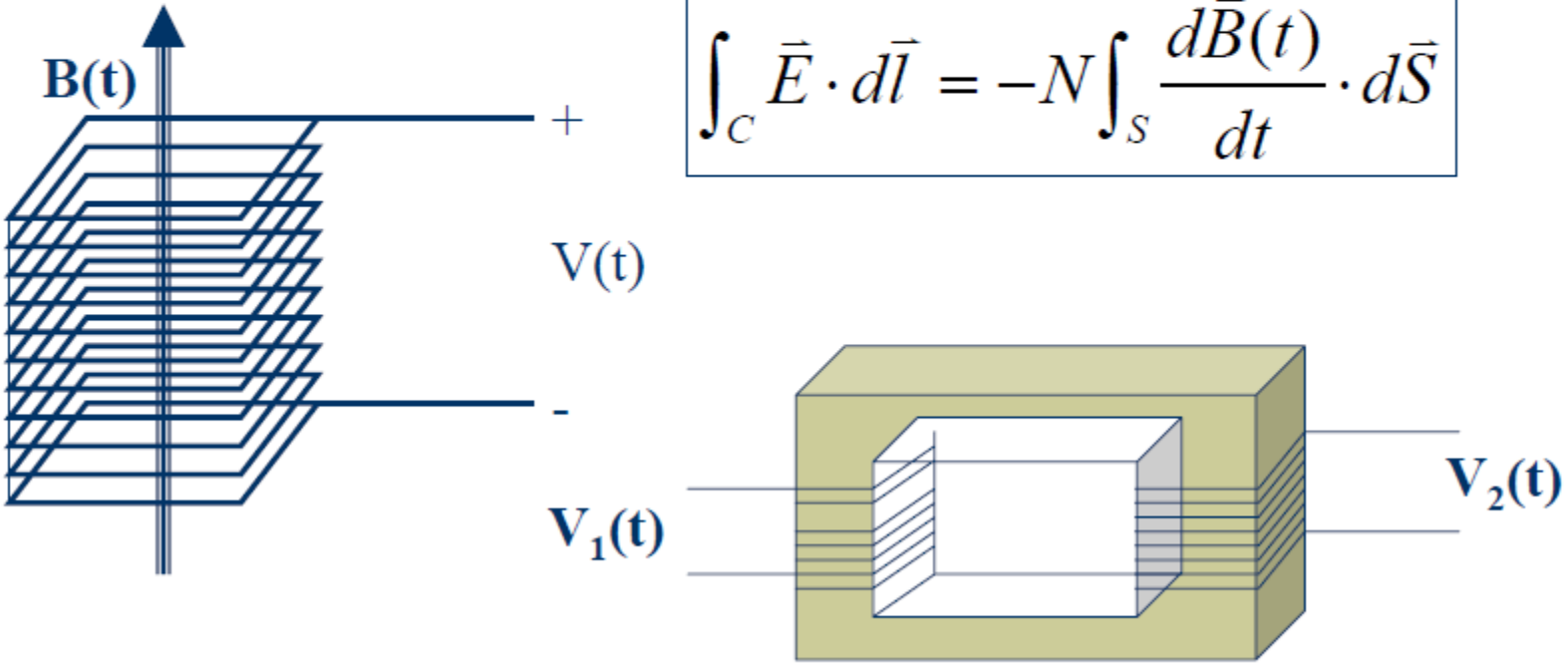
1. Changing magnetic flux density with time
2. Changing loop position with time
3. Changing both 1 and 2

$$\int_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}$$

$$= -\int_S \frac{d\vec{B}(t)}{dt} \cdot d\vec{S} - \frac{d}{dt}\int_{S(t)} \vec{B} \cdot d\vec{S}(t)$$

Emf due to the changing magnetic flux density with time: Transformer Emf

Transformer action



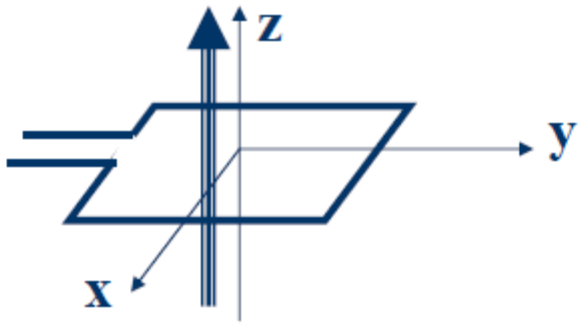
$$\int_C \vec{E} \cdot d\vec{l} = -N \int_S \frac{d\vec{B}(t)}{dt} \cdot d\vec{S}$$

Application: transformer

A stationary loop is placed such that its plane is perpendicular to the magnetic flux density.Assume the flux density is sinusoidal and given as follows:

$$\vec{B} = 0.1\sin(100\pi t)\vec{a}_z,\ \text{a} = 0.1\ \text{m},\ \text{b} = 0.1\ \text{m}$$

- Calculate the induced emf in the loop if the loop is made of N = 100 turns.



Solution

Start from Faraday's law

$$\int_C \vec{E} \cdot d\vec{l} = -N \int_S \frac{d\vec{B}(t)}{dt} \cdot d\vec{S}$$

$$emf = -100 \int_S \left(\frac{d}{dt} 0.1 \sin(100\pi t)\right) \vec{a}_z \cdot dx dy \vec{a}_z$$

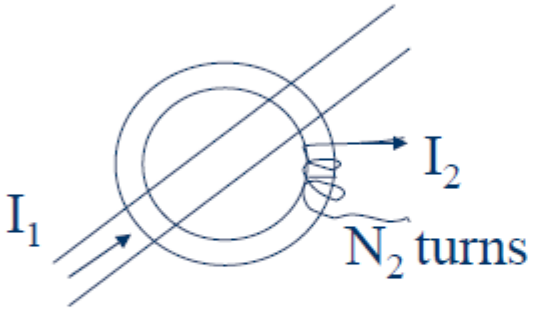
$$= -100 \times 0.1 \times 100\pi \cos(100\pi t) \int_S dx dy$$

$$= -1000\pi \cos(100\pi t)(0.1 \times 0.1)$$

$$= -10\pi \cos(100\pi t) \quad \text{V}$$

ANS

Example: The current transformer



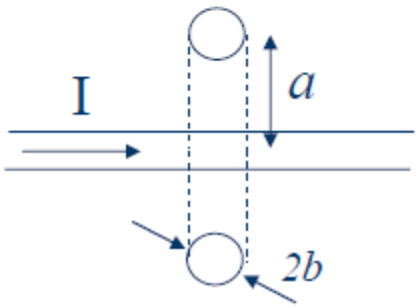
It is required to design a current transformer that will mesure sinusoidal current in the line. The peak current expected is 100 A at 60 Hz. A toroidal core, made of iron with average radius a = 30 mm. The cross section of toroid is circular with radius b = 10 mm. Relative permeability of the core is 200. The volt meter can measure between V = 0 and V = 1 V(peak).

Calculate the number of turns of toroid for full scale reading at 100 A.

$\mu_0 = 4\pi \times 10^{-7}$ H/m

1. The magnetic flux density at a distance a from the current line at full scale current is

$$B = \frac{\mu_0 \mu_r I \sin \omega t}{2\pi a} = \frac{(4\pi \times 10^{-7})200(100)\sin \omega t}{2\pi(30 \times 10^{-3})} = 0.1334 \sin(120\pi t) \quad \text{(T)}$$



2. Assuming the flux density be uniform in the core of toroid, calculate the flux

$$\Phi = BS = 0.1334 \sin(120\pi t) \times \pi \times 0.01^2$$
$$= 4.188 \times 10^{-5} \sin(120\pi t)$$

3. Calculate the induced emf in the loop

$$emf = -N_2 \frac{d\Phi}{dt} = -N_2 \left(0.0158 \cos(120\pi t)\right)$$

4. Let the maximum induced emf be 1 V, we can calculate the number of turns as

$$N_2 = \frac{|emf|_{\max}}{d\Phi/dt} = \frac{1}{d\Phi/dt} \approx 63.32 \text{ turns}$$

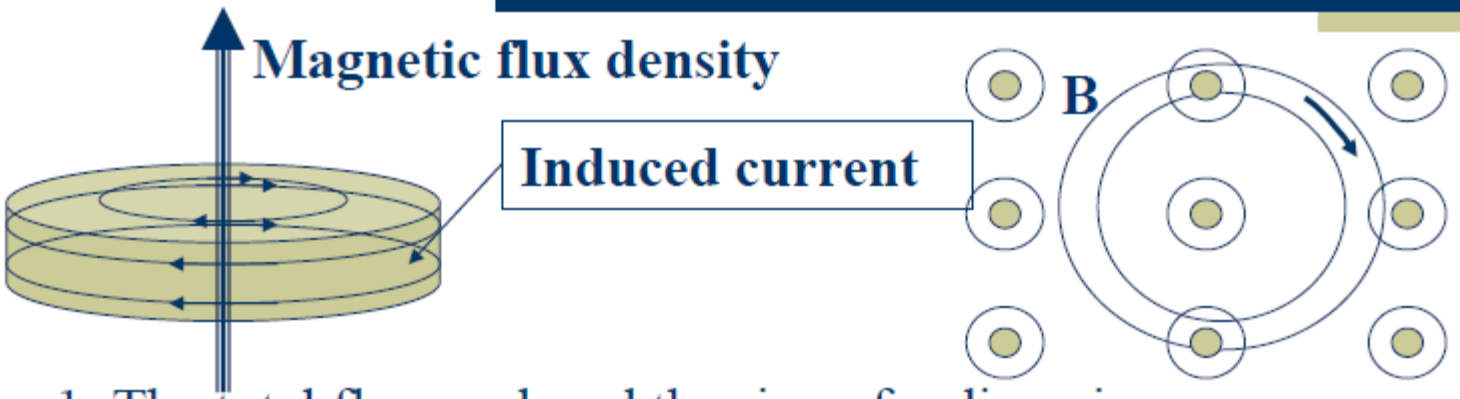
Example: Loss in conducting material due to eddy current

A circular disk of radius d = 100 mm and thickness c = 1 mm is placed in a uniform ax magnetic field. The magnetic flux density is directed perpendicular to the disk. Let us given

$$B = 0.2\sin(100\pi t) \ \text{T}, \quad \sigma = 10^7 \ \text{S/m}$$

Calculate the power dissipated in the disk due to induced current. Assume the magnetic field is not modified by the induced current.

Solution



1. The total flux enclosed the ring of radius r is

$$\Phi = \pi r^2 B_0 \sin(\omega t) \quad \text{(Wb)}$$

2. The emf (neclecting the sign since only power is need)

$$\left|\frac{d\Phi}{dt}\right| = \omega \pi r^2 B_0 \cos \omega t = emf$$

3. To calculate power we need the resistance of the ring

From Eq. 5-27 (p.204), $R = \frac{l}{\sigma S} = \frac{2\pi r}{\sigma c dr}$

4. The instantaneous power dissipate in the infinitestimal ring is

$$dP(t) = \frac{V^2}{R} = \frac{emf^2}{R} = \frac{\omega^2 \pi r^3 B_0^2 \sigma c (\cos(\omega t))^2 dr}{2} \quad \text{(W)}$$

5. Since the disl is made of infinite number of rings varying radius from zero to *d*, hence we integrate the power and get

$$P(t) = \int_0^d \frac{\omega^2 \pi r^3 B_0^2 \sigma c (\cos(\omega t))^2 dr}{2} = \frac{\omega^2 \pi d^4 B_0^2 \sigma c (\cos(\omega t))^2}{8} \text{ (W)}$$

6. Substitute the numeric value, then get the answer.

7.3 Maxwell's Equations

- Modified Ampere's law for time-varying field
- Displacement current
- Postulates of electromagnetic field for general case
- Maxwell's equations
- Boundary conditions at the interface between medium

Modification of Ampere's law for time-varying field

Start with

Electrostatic field

$$\nabla \times \vec{E} = 0$$
$$\nabla \cdot \vec{D} = \rho$$
$$\vec{D} = \varepsilon \vec{E}$$

Magnetostatic field

$$\nabla \times \vec{H} = \vec{J}$$
$$\nabla \cdot \vec{B} = 0$$
$$\vec{B} = \mu \vec{H}$$

Time-varying magnetic field

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Are there any conflicts between these 3 sets of equations ?

Check the agreement among 3 sets of equations

Start with **Magnetostatic field**

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \cdot \vec{B} = 0$$

$$\vec{B} = \mu \vec{H}$$

Take divergence

$$\nabla \cdot \nabla \times \vec{H} = \nabla \cdot \vec{J}$$

Vector identity

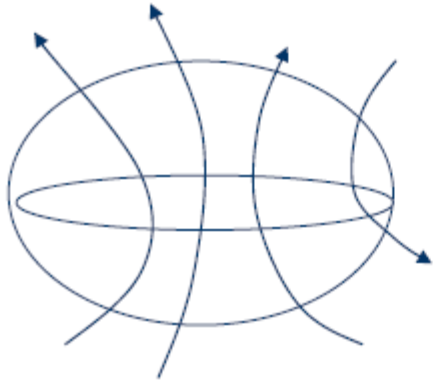
$$\nabla \cdot \nabla \times \vec{A} = 0$$

Hence

$$\nabla \cdot \vec{J} = 0$$

Is it correct?

Principle of conservation of charge



Current flow out from a closed surface

=

Rate of decresing of charge inside the closed surface

$$\oint_c \vec{J} \cdot d\vec{S} = -\frac{\partial}{\partial t}\int_v \rho dv$$

Using divergence theorem

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

Definition of divergence

$$\nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

Conflict of Ampere's law and charge conservation principle

$$\nabla \times \vec{H} = \vec{J}$$
$$\nabla \cdot \vec{B} = 0$$
$$\vec{B} = \mu \vec{H}$$

$$\Rightarrow \quad \nabla \cdot \nabla \times \vec{H} = \nabla \cdot \vec{J} \quad \Rightarrow \quad \nabla \cdot \vec{J} = 0$$ ✗

Maxwell solved this problem.

He added a new term into Ampere's law.

$$\nabla \cdot \nabla \times \vec{H} \boxed{-\frac{\partial \rho}{\partial t}} = \nabla \cdot \vec{J}$$

Electrostatic field

$$\nabla \times \vec{E} = 0$$
$$\nabla \cdot \vec{D} = \rho$$
$$\vec{D} = \varepsilon \vec{E}$$

$$\nabla \cdot \nabla \times \vec{H} - \frac{\partial \nabla \cdot \vec{D}}{\partial t} = \nabla \cdot \vec{J}$$

Hence $\Rightarrow$ $\boxed{\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}}$ New equation

Equations for time-varying field in differential form

Magnetostatic field		Time-varying field
$\nabla \times \vec{H} = \vec{J}$	→	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
$\nabla \cdot \vec{B} = 0$	→	$\nabla \cdot \vec{B} = 0$
$\vec{B} = \mu \vec{H}$	→	$\vec{B} = \mu \vec{H}$

Electrostatic field		
$\nabla \times \vec{E} = 0$	Faraday's law →	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
$\nabla \cdot \vec{D} = \rho$	→	$\nabla \cdot \vec{D} = \rho$
$\vec{D} = \varepsilon \vec{E}$	→	$\vec{D} = \varepsilon \vec{E}$

Final check for agreement

Time-varying field

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \Rightarrow \nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot \vec{J} + \frac{\partial}{\partial t} \nabla \cdot \vec{D} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\Rightarrow \quad \nabla \cdot \vec{D} = \rho$$

$$\vec{B} = \mu \vec{H}$$

$$\Rightarrow \quad \nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \nabla \cdot (\nabla \times \vec{E}) = -\frac{\partial}{\partial t} \nabla \cdot \vec{B} = 0$$

$$\nabla \cdot \vec{D} = \rho$$

$$\Rightarrow \quad \nabla \cdot \vec{B} = c_0(x, y, z)$$

$$\vec{D} = \varepsilon \vec{E}$$

$$\Rightarrow \quad c_0 = 0$$

Look closely to thte term $\frac{\partial \vec{D}}{\partial t}$

Unit $\quad \frac{\partial \vec{D}}{\partial t} : \frac{\mathrm{C}}{\mathrm{m}^2} \cdot \frac{1}{s} = \frac{A}{m^2}$ Unit of current density

$\frac{\partial \vec{D}}{\partial t}$ is called displacement current density

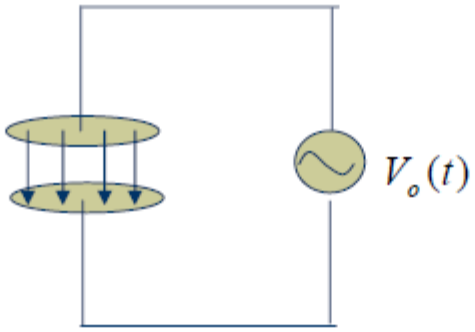
What is the nature of displacemnt current density?

Current in capacitor

Circuit theory

$$i_c = C\frac{dV_0(t)}{dt}$$

$$i_c = \varepsilon\frac{A}{d}\frac{dV_0(t)}{dt}$$



Field theory

$$V_0(t) = \int_0^d \vec{E}\cdot d\vec{l}$$

$$\frac{d}{dt}V_0(t) = \int_0^d \frac{d\vec{E}}{dt}\cdot d\vec{l}$$

$$\varepsilon\frac{d}{dt}V_0(t) = \int_0^d \frac{d\,\varepsilon\vec{E}}{dt}\cdot d\vec{l} = \int_0^d \frac{d\vec{D}}{dt}\cdot d\vec{l}$$

$$\varepsilon\frac{d}{dt}V_0(t) = \frac{d\vec{D}}{dt}d$$

$$\varepsilon A\frac{d}{dt}V_0(t) = A\frac{d\vec{D}}{dt}d$$

$$\frac{\varepsilon A}{d}\frac{d}{dt}V_0(t) = A\frac{d\vec{D}}{dt} = i_c$$

Maxwell's equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$ ⇒ Faraday's law

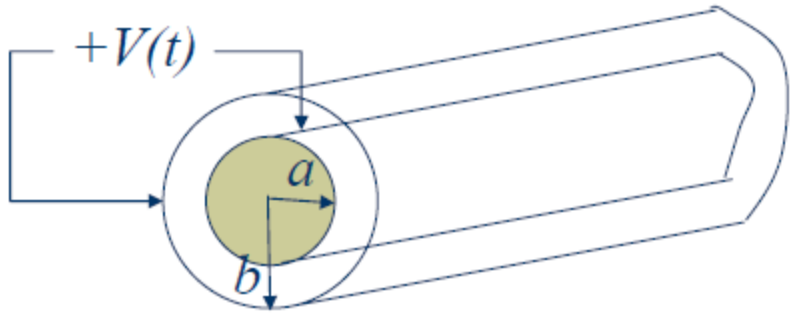
$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$ ⇒ Ampere-Maxwell's law

$$\nabla \cdot \vec{D} = \rho$$ ⇒ Gauss's law in electric field

$$\nabla \cdot \vec{B} = 0$$ ⇒ Gauss's law in magnetic field

Example of displacement current in coaxial cable

A coaxial capacitor with inner radius a = 5 mm, outer radius b = 6 mm and length 500 mm has a dielectric constant = 6.7, for which an applied voltage $V(t)$ = 250sin2π(60)t (V). Determine the displacement current.



$$\varepsilon_r = 6.7$$

$$\varepsilon_0 = \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$$

Potential function of electric field inside the coaxial cable

$$V = \left[\frac{250}{\ln(6/5)}\sin 377t\right]\left(\ln\frac{r}{0.005}\right)$$

Find electric field inside dielectric

$$\vec{E} = -\nabla V = -\frac{1.37\times10^{3}}{r}\sin 377t\ \ \vec{a}_r \quad \text{V/m}$$

$$\vec{D} = \varepsilon_0\varepsilon_r\vec{E} = -\frac{8.13\times10^{-8}}{r}\sin 377t\ \ \vec{a}_r \quad \text{C/m}^2$$

Find displacement current

$$\vec{J}_d = \frac{\partial\vec{D}}{\partial t} = -\frac{3.07\times10^{-5}}{r}\cos 377t\ \ \vec{a}_r \quad \text{A/m}^2$$

$$i_d = J_d(2\pi rL) = 9.63\times10^{-5}\cos 377t \quad \text{A}$$

$$\nabla V = \vec{a}_r\frac{\partial V}{\partial r} + \vec{a}_\phi\frac{1}{r}\frac{\partial V}{\partial \phi} + \vec{a}_z\frac{\partial V}{\partial z}$$

Composition of Maxwell's equations

- Gauss's law for electric field
- Gauss's law for magnetic field
- Faraday's law
- Ampere-Maxwell's law

Mathematical Expression of Maxwell's equations in differential form

1. Gauss's law for electric field

$$\nabla \cdot \varepsilon \vec{E} \equiv \lim_{\Delta v \to 0} \left[\frac{\oint_{\Delta S} \varepsilon \vec{E} \cdot d\vec{S}}{\Delta v} \right]$$

$$= \frac{\varepsilon}{h_1 h_2 h_3} \left[\frac{\partial (E_1 h_2 h_3)}{\partial u_1} + \frac{\partial (E_2 h_1 h_3)}{\partial u_2} + \frac{\partial (E_3 h_1 h_2)}{\partial u_3} \right] = \rho$$

Mathematical Expression of Maxwell's equations in differential form

2. Gauss's law for magnetic field

$$\nabla \cdot \vec{B} \equiv \lim_{\Delta v \to 0} \left[\frac{\oint_{\Delta S} \vec{B} \cdot d\vec{S}}{\Delta v} \right]$$

$$= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (B_1 h_2 h_3)}{\partial u_1} + \frac{\partial (B_2 h_1 h_3)}{\partial u_2} + \frac{\partial (B_3 h_1 h_2)}{\partial u_3} \right] = 0$$

Mathematical Expression of Maxwell's equations in differential form

3. Faraday's law

$$\nabla \times \vec{E} \equiv \sum_{k=u_1,u_2,u_3} \vec{a}_k \underset{\Delta s(k)\to 0}{Lim} \left[\frac{\oint_{C(k)} \vec{E} \cdot d\vec{l}}{\Delta s(k)} \right]$$

$$= \begin{vmatrix} \frac{\vec{a}_1}{h_2 h_3} & \frac{\vec{a}_2}{h_1 h_3} & \frac{\vec{a}_3}{h_1 h_2} \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 E_1 & h_2 E_2 & h_3 E_3 \end{vmatrix} = -\frac{\partial \vec{B}}{\partial t}$$

Mathematical Expression of Maxwell's equations in differential form

4. Ampere-Maxwell's law

$$\nabla \times \vec{H} \equiv \sum_{k=u_1,u_2,u_3} \vec{a}_k \lim_{\Delta s(k)\to 0} \left[\frac{\oint_{C(k)} \vec{H} \cdot d\vec{l}}{\Delta s(k)} \right]$$

$$= \begin{vmatrix} \frac{\vec{a}_1}{h_2 h_3} & \frac{\vec{a}_2}{h_1 h_3} & \frac{\vec{a}_3}{h_1 h_2} \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 H_1 & h_2 H_2 & h_3 H_3 \end{vmatrix} = \vec{J} + \frac{\partial \varepsilon \vec{E}}{\partial t}$$

Orthogonal Coordinates System

Rectangular system

Coordinate s : $u_1 = x, \quad u_2 = y, \quad u_3 = z$

Metric coefficien t : $h_1 = 1, \quad h_2 = 1, \quad h_3 = 1$

Cylindrical system

Coordinate s : $u_1 = r, \quad u_2 = \phi, \quad u_3 = z$

Metric coefficien t : $h_1 = 1, \quad h_2 = r, \quad h_3 = 1$

Spherical system

Coordinates : $u_1 = R, \quad u_2 = \theta, \quad u_3 = \phi$

Metric coefficient : $h_1 = 1, \quad h_2 = R, \quad h_3 = R\sin\theta$

Example 1

If the magnetic flux density in a region of free space is given by

$$\vec{B} = B_0 z \cos(\omega t)\vec{a}_y \quad (\mathrm{T})$$

and it is known that the time-varying electric field associated with it has only the x component.

1. Find electric field intensity vector.
2. Using Ampere's law to obtain magnetic field intensity from electric field intensity, and compare with the magnetic flux density given above.

Example 1(Soln.)

1. Using Faraday's law to obtain electric field

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ E_x & 0 & 0 \end{vmatrix} = -\frac{\partial \vec{B}}{\partial t} = \vec{a}_y \omega B_0 z \sin \omega t$$

$$\left(\frac{\partial E_x}{\partial z} \right) \vec{a}_y = \vec{a}_y \omega B_0 z \sin \omega t$$

$$\therefore E_x = \omega B_0 \frac{z^2}{2} \sin \omega t + c$$

If we use initial condition that E = 0 at t = 0, then c = 0

$$\nabla \times \vec{A} = \vec{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \vec{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \vec{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

Example 1(Soln.)

2. Using Ampere's law to obtain magnetic field

$$\nabla \times \frac{\bar{B}}{\mu_0} = \frac{1}{\mu_0}\begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & B_y & 0 \end{vmatrix} = \varepsilon_0 \frac{\partial \bar{E}}{\partial t} = \bar{a}_x \omega^2 B_0 \frac{z^2}{2} \cos \omega t$$

$$-\frac{1}{\mu_0}\left(\frac{\partial B_y}{\partial z}\right) = \omega^2 B_0 \frac{z^2}{2} \cos \omega t$$

$$\therefore B_y = -\varepsilon_0 \mu_0 \omega^2 B_0 \frac{z^3}{6} \cos \omega t$$

We can see that it is not equal to the given magnetic flux density.

$$\nabla \times \bar{A} = \bar{a}_x\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \bar{a}_y\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \bar{a}_z\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

Example 1(Soln.)

Conclusion : The obtained expression for magnetic field in part (2) is different from the originally given one. This means that the given expression of magnetic field is not a solution of the Maxwell's equations. The solution must satisfy all Maxwell's equations ***simultaneously***.

Example 2

Show that the Gauss's law can be derived from Faraday's law and Ampere's law along with continuity equation

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$

Example 2 (Soln.)

$$\nabla \cdot \nabla \times \vec{E} = 0 = \nabla \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) = -\frac{\partial \nabla \cdot \vec{B}}{\partial t}$$

Hence $\nabla \cdot \vec{B} = 0$ Gauss's law for magnetic field

$$\nabla \cdot \nabla \times \vec{H} = 0 = \nabla \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = \nabla \cdot \vec{J} + \frac{\partial \nabla \cdot \vec{D}}{\partial t}$$

As $\nabla \cdot \vec{D} = \rho$ Gauss's law for electric field

Hence $\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$ Continuity equation Q.E.D

Problems

Are these fields valid solutions of Maxwell's equations in source region?

(a) $\vec{E} = \vec{a}_y E_0 \sin(x)\sin(\omega t)$ (V/m)

(b) $\vec{B} = \vec{a}_z \dfrac{E_0}{\omega}\cos(x)\cos(\omega t)$ (T)

(c) $\vec{E} = \vec{a}_x E_0 z^2 \cos(\omega t)$

Example 7-3/4 page 1

Given a function of electric field intensity which varies with space and time as follows

$$\vec{E} = \vec{a}_x 10 \sin(\omega t - k_y y - k_z z)$$

Prove that this electric field can exists in free space region without source

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Prove that this electric field can exists in free space region without source

$$\vec{E} = \bar{a}_x 10\sin(\omega t - k_y y - k_z z)$$

First, test by using Gauss's law $\nabla \cdot \varepsilon_0 \vec{E} = 0$

$$\nabla \cdot \vec{E} = \left(\frac{\partial E_x}{\partial x}\right) + \left(\frac{\partial E_y}{\partial y}\right) + \left(\frac{\partial E_z}{\partial z}\right) = 0$$

$$\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} = \frac{\partial}{\partial x}\left(10\sin(\omega t - k_y y - k_z z\right) = 0$$